Sumanta Manna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.92
Baranagar Narendranath Vidyamandir	Class XII (West Bengal Council of Higher Secondary Education)	2020	82.80%
Baranagar Ramakrishna Mission Ashrama High School	Class X (West Bengal Board of Secondary Education)	2018	86.14%

EXPERIENCE

Research Assistant -- Brain Lab, IIT Hyderabad

- Advanced the lab's continual learning research, applying associative memory, variational inference, and Bayesian machine learning techniques to improve long-term knowledge retention across projects.
- Researched mechanistic interpretability techniques for multimodal LLMs in collaboration with industry partners, advancing continual learning capabilities across MLLM architectures.

Machine Learning Intern -- Rayoscan AI Diagnostics

- Built an RVG caries detection model, fine-tuning a U-Net from a MedSAM-based checkpoint, to improve detection accuracy by 20% and reduce false positives by 25% over the prior model.
- Deployed the model for clinical pilot testing, configuring live access across 7 clinics and ~10 doctors, to run structured A/B evaluation on 1,500+ radiographs.
- Reduced end-to-end inference latency by 10%, restructuring the request pipeline around the batched, on-demand GPU setup.

PROJECTS

Mechanistic Interpretability-Guided Continual Learning in MLLMs *(Thesis)*

- Developed attribution-patching-based circuit discovery (EAP-IG, AtP*) for complex vision-language tasks, extending a technique previously validated only on simple LLM tasks (IOI, arithmetic, capital-country lookup) to multimodal reasoning on LLaVA models.
- Designed a multimodal corruption method for attribution patching, filling a gap where no prior EAP-IG work supported complex multimodal inputs, to enable reliable circuit identification across vision and language streams.
- Matched the accuracy of MoE-style methods that store separate adapters per task, while retaining zero additional parameters post-training, demonstrating a more storage-efficient path to multi-task continual learning.

Resume-Aware Job Application Agent with Hallucination-Detection Eval Layer

- Built an end-to-end agent pipeline, scraping job postings from public Greenhouse/Lever APIs and ranking them via resume-JD embedding similarity, to surface high-fit postings.
- Designed a claim-verification layer, combining NLI-based entailment checking and LLM-judge scoring to flag unsupported claims in generated application content before human review.
- Measured the eval layer's reliability against a hand-labeled test set of injected false claims, achieving 65% precision and 85% recall in catching hallucinated content.

Mini LLM Inference Engine

- Built a custom LLM inference engine in raw PyTorch, implementing manual KV-caching and static/continuous batching, to understand and reproduce the core mechanics behind production serving systems like vLLM.
- Implemented continuous batching from scratch, replacing static batch scheduling, achieving a **4x throughput improvement** under variable-length concurrent workloads.
- Verified correctness against HuggingFace's reference implementation, matching greedy-decoded token outputs exactly across varied prompt lengths, to establish a reliable baseline before optimizing for speed.

Physically-Motivated Compound-Degradation Image Restoration -- KLA AI Hackathon

- Adapted the NAFNet architecture (NAFNetH2SR), handling joint denoising, deblurring, and 2× super-resolution, to restore radiographic-style images under compound degradation.
- Modeled the degradation pipeline empirically, using AICc-based hypothesis testing on speckle, AWGN, and downsampling noise, to guide architecture and loss design.
- Introduced a heteroscedastic Data Consistency loss (Student-t formulation) with curriculum-based OOD augmentation, achieving a Top-10 finish on the private leaderboard (**2nd place** overall) via test-time augmentation and sample-specific Adam refinement.

RELEVANT COURSES

Foundations of Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks

SKILLS

Continual Learning, NLP, Computer Vision, PyTorch, NumPy, Pandas, C/C++, Bash, SQL, REST APIs, Linux, LLM Finetuning

SCHOLASTIC ACHIEVEMENTS

Recipient of the IndiaAI fellowship by MeitY.

2nd position at KLA X IIT Hyderabad hackathon

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Deep Learning, Foundations of Machine Learning, IITH

PUBLICATIONS

Deep Learning-based Prediction and Monitoring of Parkinson's Disease using Voice Data

- Developed a custom ResNet-18 model for Parkinson's Disease detection from voice-derived acoustic features (jitter, shimmer, pitch, intensity), achieving 98% accuracy, 0.98 precision, and 0.98 F1-score, outperforming 10 baseline ML/ensemble models.

EXTRACURRICULAR

Member, State Cycling Team, West Bengal